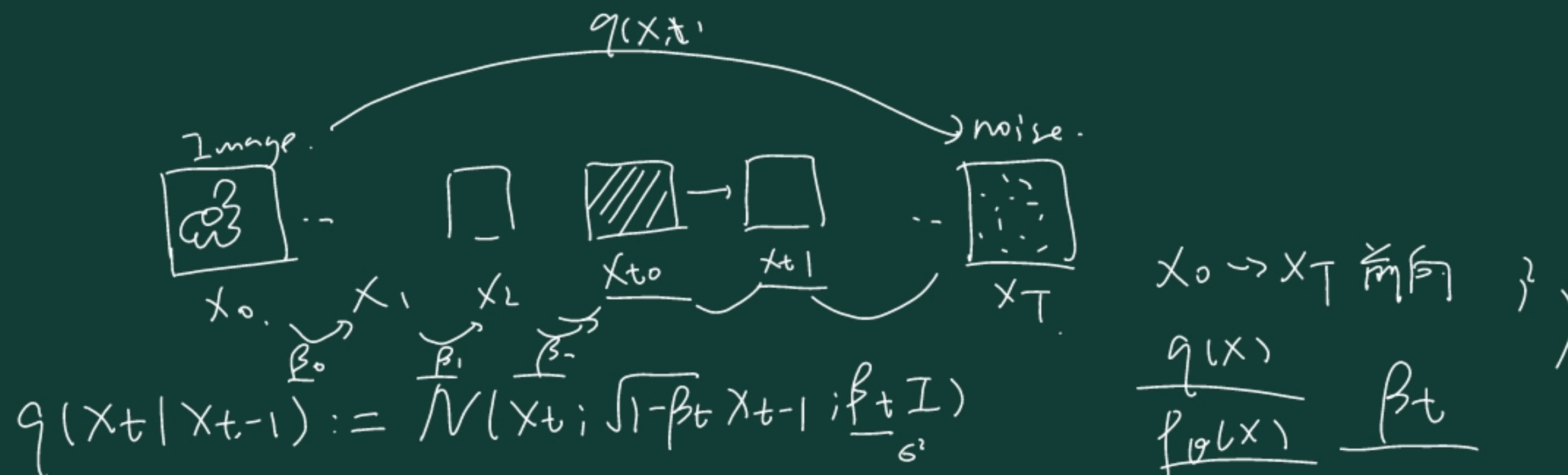




$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t} z_1 \quad \begin{matrix} z_1 \sim \mathcal{N}(0,1) \\ z_2 \sim \mathcal{N}(0,1) \end{matrix} \quad \alpha_t = 1 - \beta_t$$

$$= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1-\alpha_{t-1}} z_2) + \sqrt{1-\alpha_t} z_1 \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$$

$$= \sqrt{\alpha_t \cdot \alpha_{t-1}} x_{t-2} + \underbrace{\sqrt{1-\alpha_t} z_1 + \sqrt{\alpha_t} \cdot \sqrt{1-\alpha_{t-1}} z_2}_{\bar{\alpha}_{t-1}} = \frac{\alpha_t \cdot \alpha_{t-1} \cdot \dots \cdot \alpha_1}{\bar{\alpha}_{t-1}}$$

$$= \sqrt{\alpha_t \cdot \alpha_{t-1}} x_{t-2} + \sqrt{1-\alpha_t \cdot \alpha_{t-1}} \bar{z}_2$$

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \bar{z}_t \quad x_t = q(x_0, t)$$

Image

def q_x (x_0, t, noise)

$$\frac{\alpha_t \cdot x_0 + (1 - \alpha_t) \cdot \text{noise}}{\sqrt{\alpha_t}}$$

$$\frac{\alpha_t \cdot x_0 + (1 - \alpha_t) \cdot \text{noise}}{\sqrt{1 - \alpha_t}}$$

$$q(x_t | x_{t-1})$$

↑

$$X_t = \sqrt{\alpha_t} X_0 + \sqrt{1 - \alpha_t} \cdot Z$$

$$Z \sim N(0, 1)$$

$$\bar{Z} \sim N(0, 1)$$

$$\beta_t$$

$$\sqrt{\alpha} X_0 + \sqrt{1 - \alpha} \bar{Z} \rightarrow Z_1, Z_2$$

$$\uparrow \quad \rightarrow 0$$

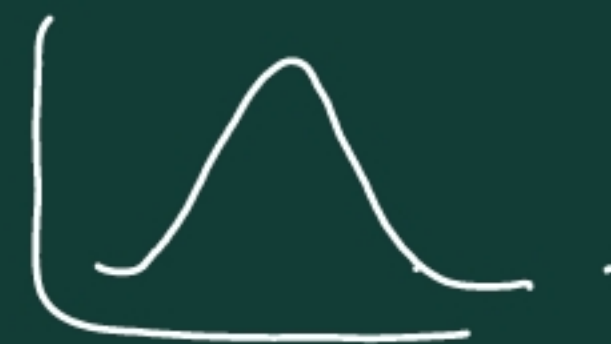
$$\uparrow \quad \rightarrow 1$$

$$\alpha_t \rightarrow 0$$

$$\alpha_t \cdot \alpha_{t-1} \dots \alpha_1 \rightarrow 0$$

$$(1 - \beta_t)(1 - \beta_{t+1}) \dots (1 - \beta_1) \rightarrow 0$$

$$\frac{\beta_1}{1 - \beta_1}$$



$$\alpha_t = 1 - \beta_t$$

$$\alpha_t = \frac{1}{1 + t} \alpha_s$$

$$\beta_t \quad T = 100$$

$$1 \times 10^{-5} \rightarrow 5 \times 10^{-3}$$

$$\beta_1 = 1 \times 10^{-5}$$

$$\beta_{100} = 5 \times 10^{-3}$$

noise-estimation-loss

$$\sqrt{(e - \text{output})^2}$$

model(x, t)

$$x_0 \cdot \alpha + e \cdot \alpha_m$$

$$\sqrt{\alpha_t}$$

$$\sqrt{1 - \alpha_t}$$

$$\rightarrow \sqrt{(e - \text{model}) (\sqrt{\alpha_t} \cdot x_0 + \sqrt{1 - \alpha_t} \cdot e)^2}$$

no

torch.randn-like

p-sample

x_{t-1}

$$\text{mean} + \underline{\text{sigma}_t} \times z.$$

|

β_t

$$\frac{1}{\sqrt{\alpha_t}} \times (x - (\text{eps_factor} \times \text{eps_theta}_t))$$

$$\frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}}$$

$$\frac{\text{model}(x, t)}{\Sigma_\theta(x, t)}$$